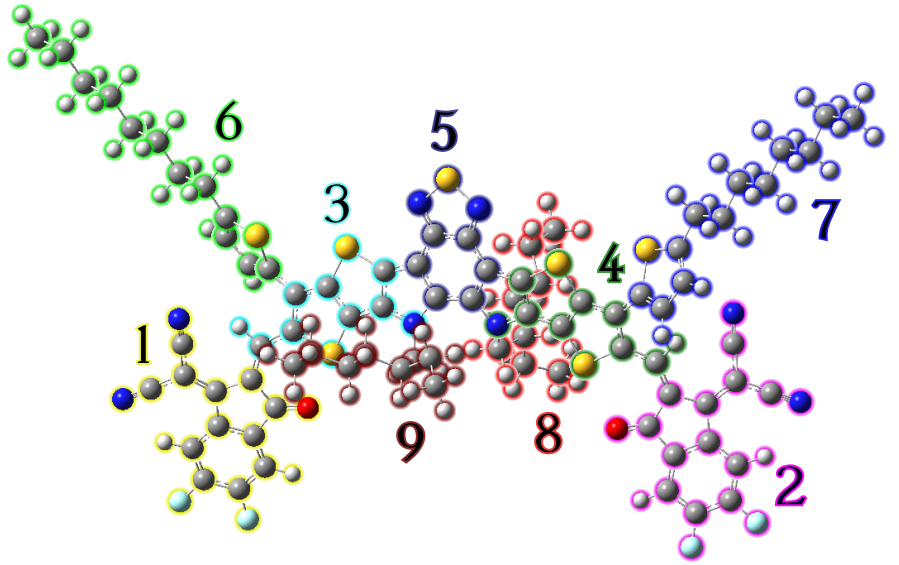
a)



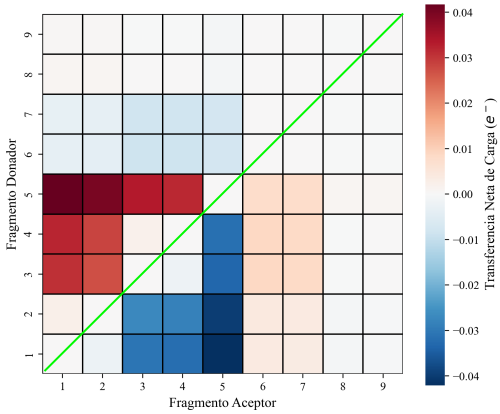
b)

HOMO→LUMO 100%

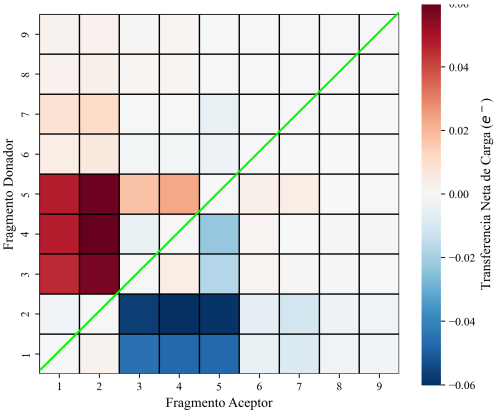
HOMO→LUMO 70%
HOMO-1→LUMO+1 25%

HOMO→LUMO 95%

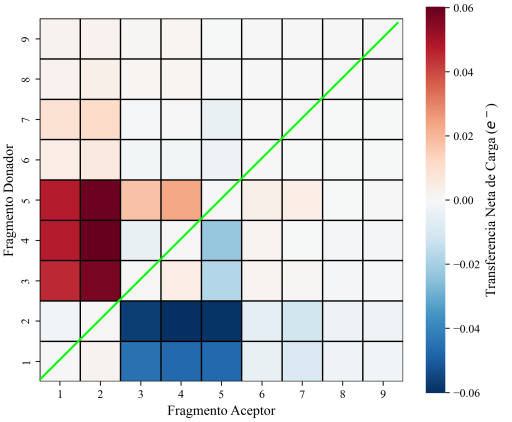
1 EE



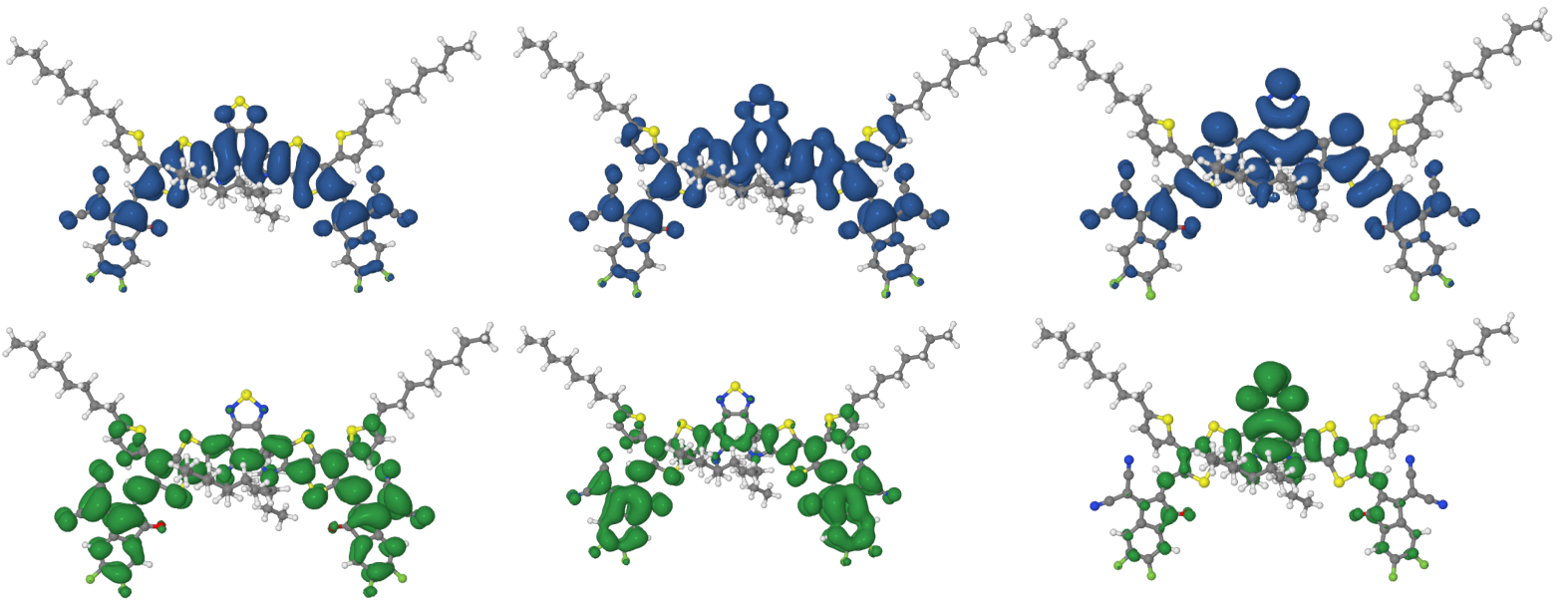
8 EE



39 EE



c)



$D_{index} = 1.203$ Å

$D_{index} = 2.430$ Å

$D_{index} = 2.434$ Å